Molarity and Solution Stoichiometry

Molarity, making and diluting solutions, and titrations

Name _____ Class _____ Date _____

DATA

H = 1.008 g/mol O = 16.00 g/mol Na = 22.99 g/mol Cl = 35.45 g/mol K = 39.10 g/mol

Molarity = moles ÷ litres of solution. Remember: mL ÷ 1000 = L.

A. Understanding

1. Define molarity, and give its unit.

2. A student dissolves a solid in exactly 250.0 mL of water and assumes the solution volume is 250.0 mL. Why is this not quite right, and how should the solution be made?

B. Calculating molarity and amounts

3. 10.0 g of sodium hydroxide is dissolved to make 500.0 mL of solution. What is the molarity?

4. What mass of potassium chloride, KCl, is needed to make 250.0 mL of 0.150 M solution?

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

5. How many moles of solute are in 35.0 mL of a 0.200 M solution?

C. Dilution

6. What volume of 12.0 M HCl is needed to make 500.0 mL of 0.600 M HCl?

7. 50.0 mL of 2.00 M NaCl is diluted to 400.0 mL. What is the new concentration?

D. Titrations

8. 22.50 mL of 0.120 M NaOH exactly neutralizes 25.00 mL of hydrochloric acid. What is the concentration of the acid? $\mathrm{HCl} + \mathrm{NaOH} \longrightarrow \mathrm{NaCl} + \mathrm{H_2O}$

9. 18.75 mL of 0.200 M NaOH exactly neutralizes 15.00 mL of sulfuric acid. What is the concentration of the sulfuric acid? $\mathrm{H_2SO_4} + 2\,\mathrm{NaOH} \longrightarrow \mathrm{Na_2SO_4} + 2\,\mathrm{H_2O}$

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.